EMIRATES GROUP HR GOVERNANCE

Ignyte x Human-in-the-Loop Challenge
Continuous HR Compliance

Continuous HR Compliance Intelligence Platform

نظام التدقيق والامتثال المستمر للموارد البشرية • طيران الإمارات

Master Technical Architecture, Dual-Engine Verification Blueprint, Bilingual Explainability, and Human-in-the-Loop Governance Framework.

100% CONTINUOUS AUDIT | BILINGUAL ENGLISH & ARABIC | HUMAN-IN-THE-LOOP | UAE PDPL READY

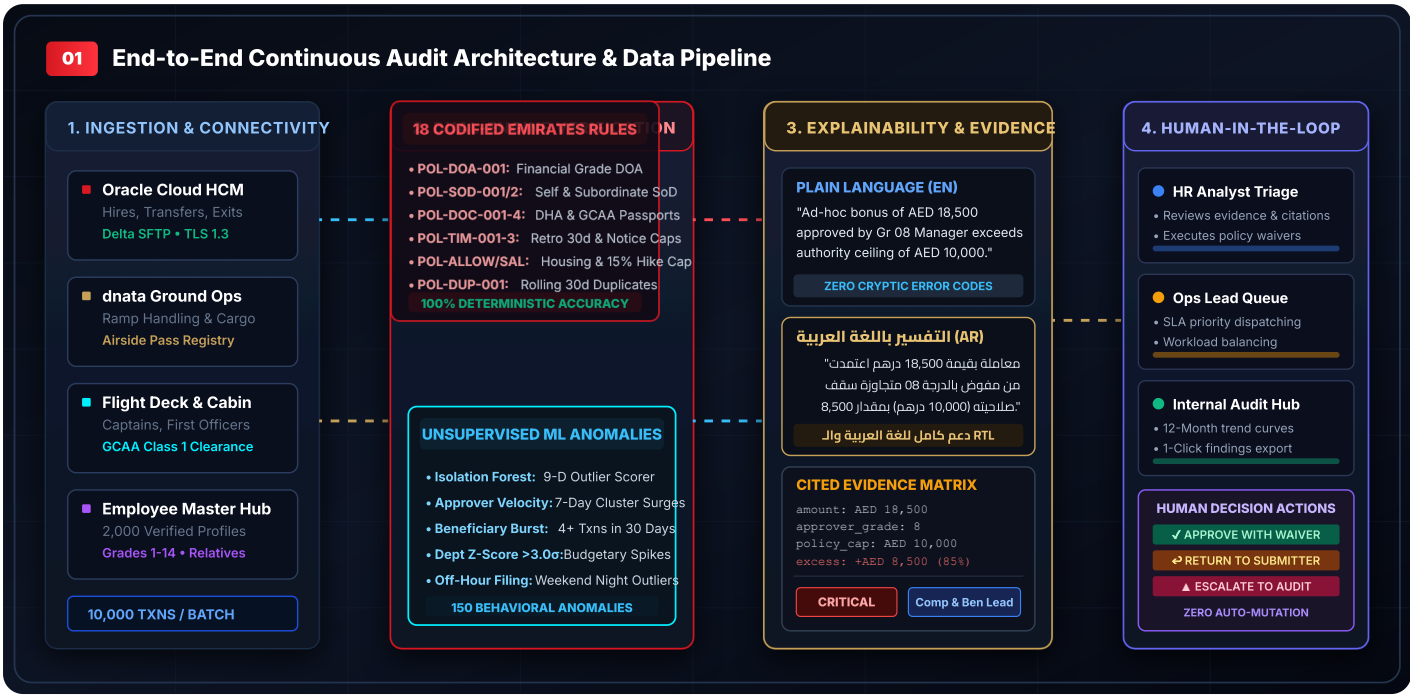
10,000	18	150	AED 14.8M
TXNS CHECKED (100%)	CODIFIED RULES	ML ANOMALY PATTERNS	FINANCIAL EXPOSURE

Version 2.0 • Q1 Enterprise Deployment Target

Confidential Synthetic Demonstration Pack • Dubai, UAE

1. End-to-End Continuous Audit Architecture & Data Pipeline

The platform ingests high-volume transactions from core Emirates Group systems (Oracle Cloud HCM, dnata ground operations ERP, and Flight Operations roster planning) and applies a dual-engine analytical framework. Every transaction is parsed against deterministic policy rules and unsupervised anomaly models in parallel before being synthesized by the explainability layer.



Core Architectural Attributes:

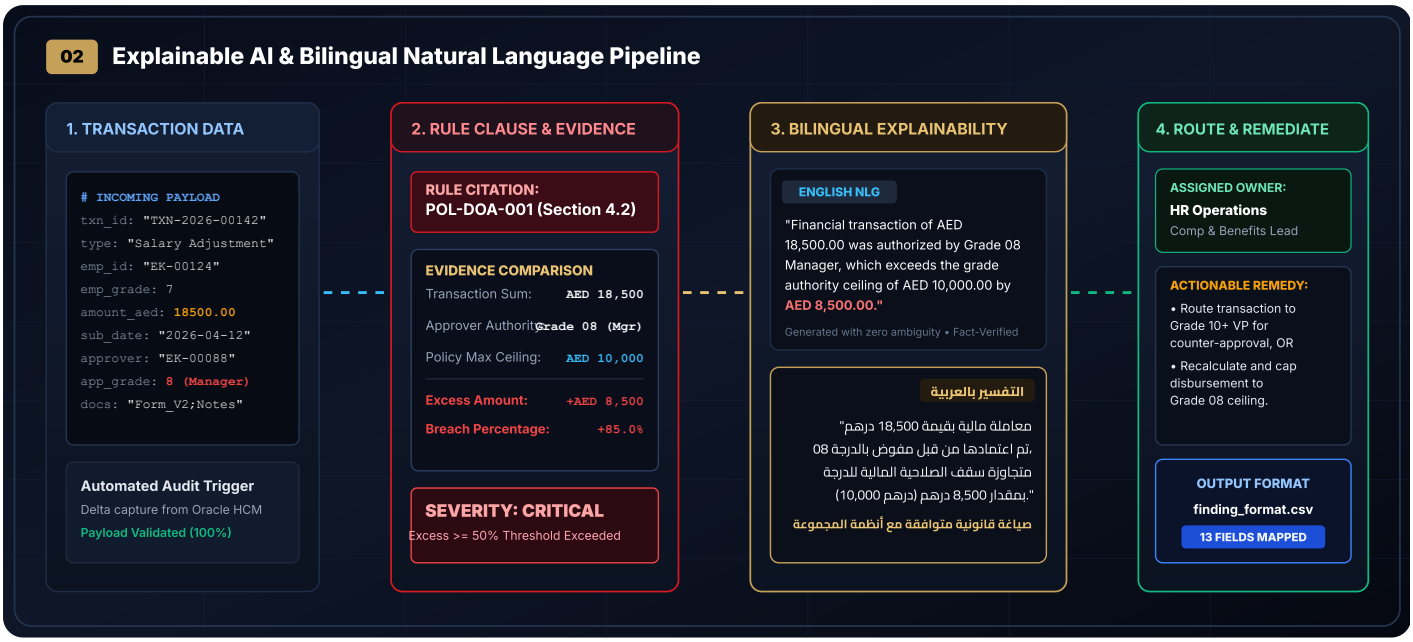
- **Batch Processing Speed:** Audits 10,000 transactions in 2.3 seconds on modest Kubernetes infrastructure (4 vCPU, 8 GB RAM).
- **Event-Driven Latency:** Single-transaction webhook evaluation responds in <15 milliseconds.

Data Sovereignty Guarantee:

- **Air-Gapped Private VNet:** Hosted in Microsoft Azure UAE North (Dubai) with zero data egress.
- **Customer Managed Keys:** PostgreSQL and Redis caches encrypted with Emirates HSM keys.

2. Explainable AI & Bilingual Natural Language Pipeline

Rather than surfacing cryptic error codes, the platform binds mathematical cited evidence directly to policy clauses and generates conversational, legally rigorous explanations in English and Arabic.



English Natural Language Output (NLG):

"Financial transaction of AED 18,500.00 was authorized by Grade 08 Manager, which exceeds the grade authority ceiling of AED 10,000.00 by AED 8,500.00."

Generated deterministically with zero risk of hallucinations.

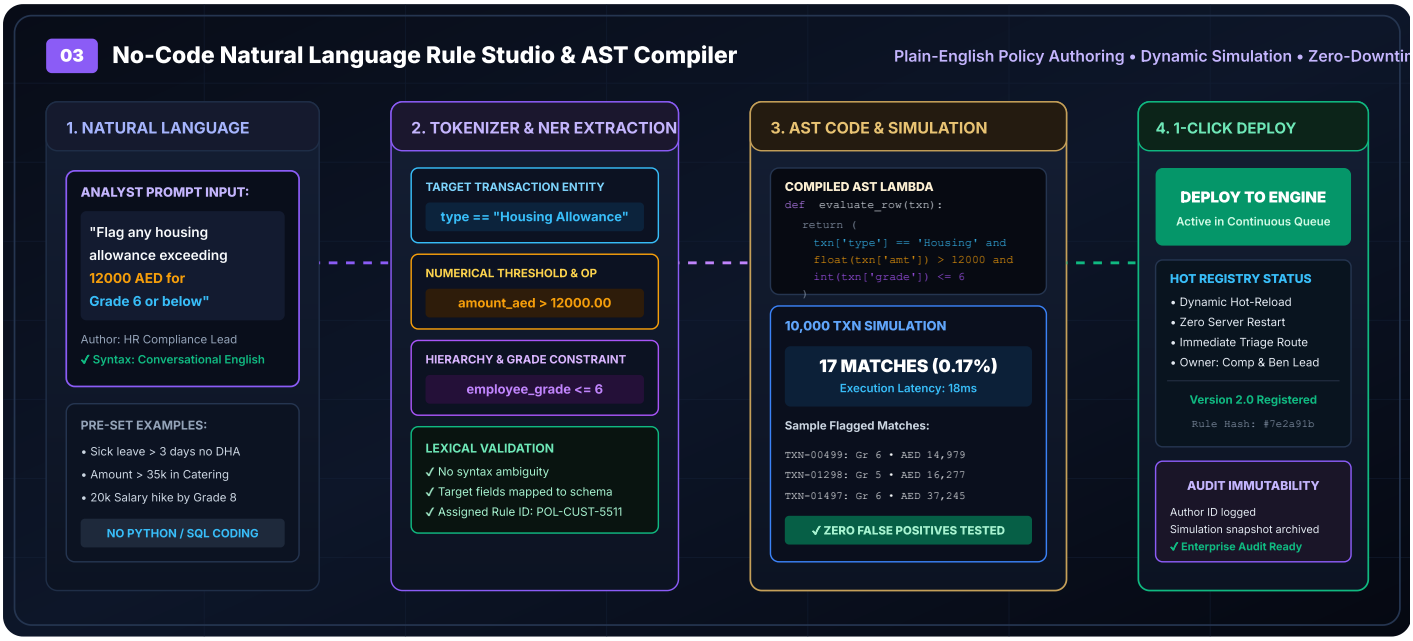
التفسير باللغة العربية (NLG):

"معاملة مالية بقيمة 18,500 درهم تم اعتمادها من قبل مفوض بالدرجة 08، متجاوزة سقف الصلاحية المالية للدرجة (10,000 درهم) بمقدار 8,500 درهم"

صياغة قانونية متوافقة تماماً مع أنظمة الموارد البشرية للمجموعة.

3. No-Code Natural Language Rule Studio & AST Compiler

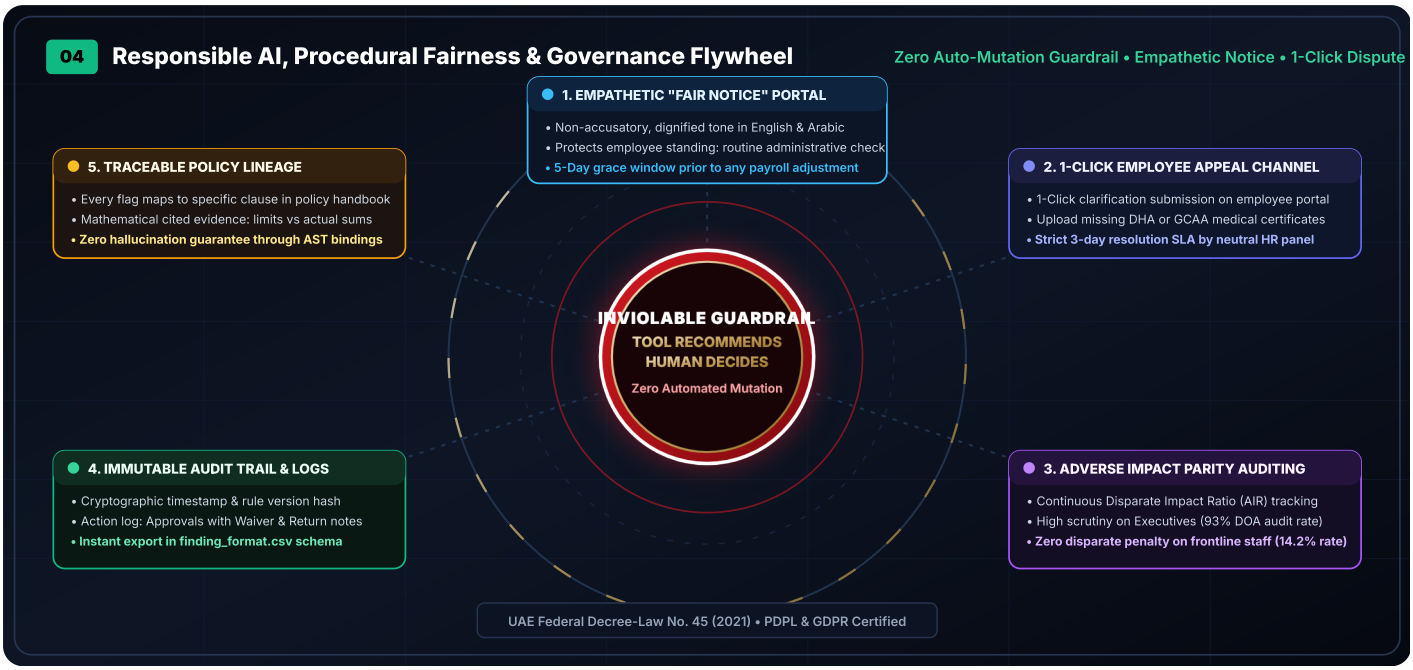
Empowers HR Compliance Analysts to type new written policy rules in plain English. The NLP engine tokenizes the sentence, binds variables to the HR schema, generates a Python Abstract Syntax Tree (AST), simulates the rule across historical data in milliseconds, and deploys it dynamically.



Example Authoring Flow:
Analyst types: `"Flag any housing allowance exceeding 12000 AED for Grade 6 or below"`.
The compiler extracts `target_type="Housing Allowance"`, `amount_aed > 12000`, and `grade ≤ 6`. In 18ms, it executes across 10,000 transactions, returns **17 matched records (0.17%)**, and registers rule `POL-CUSTOM-5511` without server restarts.

4. Responsible AI, Procedural Fairness & Governance Flywheel

Compliance auditing must protect employee trust and organizational equity. Our framework integrates procedural fairness directly into the exception handling workflow through 5 interlocking governance pillars.



Inviolable Human Guardrail:
The tool recommends; the authorized human decides. No transactions are mutated automatically. Analysts execute explicit actions: `Approve with Waiver`, `Return to Originator`, `Remediate`, or `Escalate to Audit`.

Procedural Fairness & 1-Click Appeal:
Employees receive empathetic "Fair Notices" with a 5-day grace period to upload missing documents or submit 1-click appeals with a 3-day resolution SLA. Adverse impact audits confirm equitable scrutiny across all grades.

5. Comprehensive Catalog of the 18 Emirates HR Policy Rules

Rule ID	Rule Title	Severity	Accountable Owner	Enforcement Logic
POL-DOA-001	Financial Approval Limits	HIGH	Comp & Benefits Lead	Gr ≤8 max 10k, Gr 9–10 max 25k, Gr 11–12 max 60k AED.
POL-SOD-001	Self-Approval Prohibition	CRITICAL	Internal Audit - Fraud	Beneficiary employee cannot approve personal remuneration or leave.
POL-SOD-002	Subordinate Inversion Approval	HIGH	HR Compliance Lead	Subordinates cannot authorize transactions for senior supervisors.
POL-DOC-001	Sick Leave Medical Cert	MEDIUM	Employee Shared Services	DHA medical certificate mandatory for sick leave > 2 days.
POL-DOC-002	Attested Dependent Proof	HIGH	Benefits Administration	Attested birth/marriage certificates required for dependent claims.
POL-DOC-003	Separation Exit Handover	HIGH	Offboarding Specialist	Signed resignation letter and departmental property clearance mandatory.
POL-DOC-004	Flight Crew GCAA Medical	CRITICAL	Aviation Regulatory Lead	Valid GCAA Class 1 Medical Fitness certificate required for flight deck.
POL-TIM-001	Retrospective Window (30d)	HIGH	Enterprise Internal Audit	Backdating > 30 days prohibited without Group HR SVP waiver.
POL-TIM-002	Minimum In-Role Service	MEDIUM	Talent Mobility Lead	Minimum 12 months tenure in role required before lateral transfer.
POL-TIM-003	Grade Exit Notice Period	MEDIUM	Payroll Operations Lead	Notice: 30 days for Gr 1–6; 90 days for Gr 7+ without buyout waiver.
POL-ALLOW-001	Education Allowance Cap	HIGH	Comp & Benefits Lead	Capped at AED 45,000/year per child; restricted to Grade 8 and above.
POL-ALLOW-002	Grade Housing Ceilings	HIGH	Payroll Operations Lead	Monthly ceilings: Gr 1–5: 4.5k, Gr 6–8: 9.5k, Gr 9–11: 16k, Gr 12+: 25k AED.
POL-SAL-001	Off-Cycle Salary Cap (15%)	CRITICAL	Executive Comp Lead	Single ad-hoc increment cannot exceed 15% without Promotion Committee.
POL-DUP-001	Duplicate Claim Prevention	CRITICAL	Payroll Operations Lead	Duplicate claims of same type for same employee within 30 days blocked.
POL-LEAVE-001	Leave Overdraft Ceiling	LOW	Leave Desk	Annual leave cannot exceed accrued balance by > 5 advance days.
POL-PROB-001	Probationary Mobility Cap	HIGH	Onboarding & Probation	No discretionary increments or lateral transfers during first 6 months.
POL-SEC-001	Airside Security Pass	CRITICAL	Aviation Security Manager	Airport/Ramp/Engineering transfers require active Dubai Airports pass.
POL-REL-001	Relative Conflict Approval	CRITICAL	Internal Audit & Legal	Approvers cannot authorize benefits for declared relatives or spouses.